OHM'S LAW AND KIRCHHOFF'S RULES

1. Ohm's Law

Definition: Ohm's law states that the current flowing through a conductor is directly proportional to the potential difference across it, provided temperature and other physical conditions remain constant.

Formula:

V = IR

Where,

- **V = Voltage (volts)**
- **I = Current (amperes)**
- **R = Resistance (ohms)**

Other forms:

I = V/R

R = V/I

Explanation:

When the voltage increases, the current increases if resistance remains constant. Similarly, when resistance increases, current decreases if voltage remains constant.

Limitations:

Ohm's law is applicable to ohmic conductors under constant physical conditions. It does not generally apply to non-ohmic devices such as diodes.

2. Kirchhoff's Rules

Definition: Kirchhoff's rules are used to determine unknown currents and voltages in electrical circuits. They are based on the conservation of electric charge and conservation of energy.

There are two Kirchhoff's rules:

A. Kirchhoff's Current Law (KCL)

Statement: The total current entering a junction is equal to the total current leaving the junction.

Formula:

Sum of incoming currents = Sum of outgoing currents

$I_1 = I_2 + I_3$

Explanation:

A junction is a point where two or more branches of an electrical circuit meet. According to the law of conservation of charge, electric charge cannot accumulate at a junction in a steady-state circuit.

Example:

If 10 A of current enters a junction and 4 A leaves through one branch, then the current in the other branch is:

$10 = 4 + I_3$

$I_3 = 6$ A

Applications:

1. **Used to calculate unknown branch currents.**
2. **Used to analyze parallel circuits.**
3. **Used to solve complex electrical networks.**

B. Kirchhoff's Voltage Law (KVL)

Statement: The algebraic sum of all voltages around any closed loop of an electrical circuit is zero.

Formula:

$\Sigma V = 0$

Or,

Sum of voltage rises = Sum of voltage drops

For a simple circuit containing a battery and two resistors:

$E - IR_1 - IR_2 = 0$

Therefore,

$E = IR_1 + IR_2$

Where,

- **E = Electromotive force (EMF)**
- **I = Current**
- **R_1 and R_2 = Resistances**

Explanation:

According to the law of conservation of energy, the energy supplied by a voltage source is equal to the energy consumed by the circuit components around a closed loop.

Example:

If a battery supplies 12 V and the voltage drops across two resistors are 7 V and 5 V, then:

$12 - 7 - 5 = 0$

Hence, Kirchhoff's Voltage Law is satisfied.

Sign conventions:

1. **Moving through a resistor in the direction of current gives a voltage drop of −IR.**
2. **Moving through a resistor opposite to the current gives a voltage rise of +IR.**
3. **Moving through a battery from the negative terminal to the positive terminal gives +E.**
4. **Moving through a battery from the positive terminal to the negative terminal gives −E.**

Applications:

1. **Used to calculate unknown voltages and currents.**
2. **Used to analyze closed electrical loops.**
3. **Used to solve complex circuits containing multiple voltage sources and resistors.**

3. Difference Between KCL and KVL

Kirchhoff's Current Law (KCL)	Kirchhoff's Voltage Law (KVL)
Also called the junction rule.	Also called the loop rule.
Based on conservation of charge.	Based on conservation of energy.
Applied at a junction or node.	Applied around a closed loop.
Deals with electric current.	Deals with voltage.
Incoming current equals outgoing current.	Algebraic sum of voltages is zero.

4. Conclusion

Ohm's law describes the relationship between voltage, current, and resistance. Kirchhoff's Current Law is used to calculate currents at junctions, whereas Kirchhoff's Voltage Law is used to calculate voltages around closed loops. These laws are fundamental for analyzing electrical circuits.